

Solutions to Exercises 2

Exercise 1 (Traces).

1. Since there are n variables, which can take two values, there are 2^n possible states. We can construct a trace passing through each state exactly once.
2. As in class, $\bar{s} = (s_1, \dots, s_n)$ and exponents, e.g. $\bar{s}^k$, are used to represent variables in different states (i.e. $\bar{s}^k = (s_1^k, \dots, s_n^k)$).

(a) Noting

$$\bar{v}^k = (s_1^k, \dots, s_n^k, a_1^k, \dots, a_m^k, x_1^k, \dots, x_\ell^k),$$

$$F_{(=k)} = \exists \bar{v}^0, \dots, \bar{v}^{k-1}, \bar{s}^k. \text{Init} [\bar{s} := \bar{s}^0] \wedge \left(\bigwedge_{i=0}^{k-1} R_i \right) \wedge (\bar{s}^k = \bar{s}).$$

Another (equivalent) solution:

$$\exists \bar{v}^0, \dots, \bar{v}^{k-1}.$$

$$\text{Init} [\bar{s} := \bar{s}^0] \wedge \left(\bigwedge_{i=0}^{k-2} R_i \right) \wedge R \left[(\bar{s}, \bar{a}, \bar{x}, \bar{s}') := (\bar{s}^{k-1}, \bar{a}^{k-1}, \bar{x}^{k-1}, \bar{s}) \right].$$

(b) $F_{(\leq k)} = \bigvee_{i=0}^k F_{(=i)}.$

(c) The “intuitive” solution is $F_{(\leq \infty)}$. However, this formula is infinite. Using question (1), it can be reduced to $F_{(\text{reachable})} = F_{(\leq 2^n - 1)}$.

(d) $F_{(=k, \text{Err})} = \exists \bar{v}^0, \dots, \bar{v}^{k-1}, \bar{s}^k. (\bar{s}^0 = \bar{s}) \wedge \left(\bigwedge_{i=0}^{k-1} R_i \right) \wedge E [\bar{s} := \bar{s}^k]$

(e) $F_{(\leq k, \text{Err})} = \bigvee_{i=0}^k F_{(=i, \text{Err})}$

(f) $F_{(\text{stuck})} = \neg \exists \bar{a}, \bar{x}, \bar{s}'. R$

3. The formulas are:

(a) $\neg \exists \bar{s}. F_{(\text{reachable})} \wedge E$

(b) $\neg \exists \bar{s}. F_{(\text{reachable})} \wedge F_{(\text{stuck})}$

(c) $\forall \bar{s}. F_{(\leq k)}$

- (d) Note that we are only allowed to quantify on propositional variables. As such, a formula such as

$$\forall k. \exists \bar{s}. F_{(=k)} \wedge \neg F_{(\text{stuck})}$$

is not allowed, as k ranges (implicitly) over $\mathbb{N}$.

Consider instead

$$\exists \bar{s}. F_{(=2^n)}$$

if such a state ($\bar{s}$) exists, the underlying trace contains $2^n + 1$ states (since it is 2^n steps long). By the pigeon-hole principle, one of these states must appear twice in the trace. This means that there is a part of the trace which can be repeated infinitely (“pumped”), implying the existence of an infinite trace (see the proof of the [pumping lemma](#) for regular languages which relies on the same principle).

Exercise 2 (Tseytin Transformation).

- Any expression of the type $y_k \leftrightarrow x_i * x_j$ is easily convertible to a CNF.

$$\begin{aligned} y_k \leftrightarrow x_i \vee x_j &\equiv (y_k \rightarrow x_i \vee x_j) \wedge (x_i \vee x_j \rightarrow y_k) \\ &\equiv (\neg y_k \vee x_i \vee x_j) \wedge (\neg x_i \vee y_k) \wedge (\neg x_j \vee y_k) \\ y_k \leftrightarrow x_i \wedge x_j &\equiv (y_k \rightarrow x_i \wedge x_j) \wedge (x_i \wedge x_j \rightarrow y_k) \\ &\equiv (\neg y_k \vee x_i) \wedge (\neg y_k \vee x_j) \wedge (\neg x_i \vee \neg x_j \vee y_k) \\ y_k \leftrightarrow x_i \rightarrow x_j &\equiv y_k \leftrightarrow \neg x_i \vee x_j \\ &\equiv (\neg y_k \vee \neg x_i \vee x_j) \wedge (x_i \vee y_k) \wedge (\neg x_j \vee y_k) \\ y_k \leftrightarrow \neg x_i &\equiv (y_k \rightarrow \neg x_i) \wedge (\neg x_i \rightarrow y_k) \\ &\equiv (\neg y_k \vee \neg x_i) \wedge (x_i \vee y_k) \end{aligned}$$

Let’s first define the mapping between sub formulas and fresh variables starting from the leaves:

$$\begin{aligned} y_1 &\leftrightarrow \neg b \\ y_2 &\leftrightarrow a \vee y_1 \\ y_3 &\leftrightarrow b \rightarrow c \\ y_4 &\leftrightarrow \neg y_3 \\ y_5 &\leftrightarrow y_2 \wedge y_4 \\ y_6 &\leftrightarrow y_5 \rightarrow a \\ y_7 &\leftrightarrow \neg y_6 \end{aligned}$$

The Tseytin transformation of the formula is therefore:

$$\begin{aligned}
& (y_1 \leftrightarrow \neg b) \wedge (y_2 \leftrightarrow a \vee y_1) \wedge (y_3 \leftrightarrow b \rightarrow c) \wedge (y_4 \leftrightarrow \neg y_3) \wedge \\
& (y_5 \leftrightarrow y_2 \wedge y_4) \wedge (y_6 \leftrightarrow y_5 \rightarrow a) \wedge (y_7 \leftrightarrow \neg y_6) \wedge y_7 \\
= & (\neg y_1 \vee \neg b) \wedge (b \vee y_1) \wedge (\neg y_2 \vee a \vee y_1) \wedge (\neg a \vee y_2) \wedge (\neg y_1 \vee y_2) \wedge \\
& (\neg y_3 \vee \neg b \vee c) \wedge (b \vee y_3) \wedge (\neg c \vee y_3) \wedge (\neg y_4 \vee \neg y_3) \wedge (y_3 \vee y_4) \wedge \\
& (\neg y_5 \vee y_2) \wedge (\neg y_5 \vee y_4) \wedge (\neg y_2 \vee \neg y_4 \vee y_5) \wedge (\neg y_6 \vee \neg y_5 \vee a) \wedge \\
& (y_5 \vee y_6) \wedge (\neg a \vee y_6) \wedge (\neg y_6 \vee \neg y_7) \wedge (y_6 \vee y_7) \wedge y_7
\end{aligned}$$

2. We first show that they might not be equivalent by providing a small counterexample. As written above, the Tseytin transform of the formula $\neg x$ is $(y \leftrightarrow \neg x) \wedge y$. The value of $\neg x$ when $x = 0$ is 1, whereas the value of its Tseytin transformation when $x = 0, y = 0$, is 0. Therefore the two formulas are not equivalent.

Let's now prove that a formula is satisfiable if and only if its Tseytin transformation is satisfiable as well.

Let φ be a formula and $V = \text{FV}(\varphi)$ its set of free variables. One can notice (and prove by induction) that the Tseytin transformation of φ is equivalent to:

$$y_n \wedge \bigwedge_{i=1}^n (y_i \leftrightarrow \psi_i)$$

where $\{\psi_i\}_{0 \leq i \leq n}$ are all the (non-atomic) subformulas of φ ; in particular, $\psi_n = \varphi$.

φ is satisfiable iff there exists an assignment α such that $\varphi[V := \alpha] = 1$. However, note that the formula $(y_i \leftrightarrow \psi_i)$ is true if and only if one assigns the truth value of $\psi_i[V := \alpha]$ to y_i . We can construct a new assignment α' as follows:

$$\alpha'(v) = \begin{cases} \psi_i[V := \alpha] & \text{if } v = y_i \text{ for some } 0 \leq i \leq n \\ \alpha(v) & \text{otherwise} \end{cases}$$

Convince yourself that this is a satisfying assignment for both the original formula φ and its Tseytin transformation. Conversely, let α' be a satisfying assignment for the Tseytin transformation of φ . This is already a satisfying assignment for φ .

Thus, there exists a satisfying assignment for φ if and only if there exists a satisfying assignment for its Tseytin transformation. Therefore, they are equisatisfiable.

3. F is a valid formula if and only if $\neg F$ is false for any variable assignment, i.e. it is unsatisfiable. A formula is unsatisfiable if and only if its Tseytin transformation is unsatisfiable as well (by the previous question). Therefore F is a valid formula if and only if the Tseytin transformation of $\neg F$ is unsatisfiable.

By induction on the structure of the formula, one can show that formula size increases by at most a constant amount for each "building block" of the formula (literal or binary operator). Therefore the size of the Tseytin transformation is in $O(n)$.

Exercise 3 (SAT solving). By taking the CNF found in the above exercise and applying resolution, we obtain the following derivation trees:

$$\begin{array}{c}
\frac{\frac{\frac{\neg a \vee y_6}{\neg a} \quad \frac{y_7 \quad \neg y_6 \vee \neg y_7}{\neg y_6}}{\neg a \text{ (1)}} \quad \frac{\frac{\frac{b \vee y_3}{b} \quad \frac{\neg y_3 \vee \neg y_4}{\neg y_3}}{b \text{ (2)}} \quad \frac{\frac{\frac{y_5 \vee y_6}{y_5} \quad \frac{y_7 \quad \neg y_6 \vee \neg y_7}{\neg y_6}}{y_5 \vee y_6} \quad \frac{y_4 \vee \neg y_5}{y_4}}{y_5} \\
\frac{\frac{\neg y_1 \vee \neg b \quad \neg y_2 \vee a \vee y_1}{\neg b \vee a \vee \neg y_2} \quad \frac{\neg y_5 \vee y_2}{y_2}}{\neg b \vee a \text{ (3)}} \quad \frac{\frac{b \text{ (2)}}{b} \quad \frac{\neg b \vee a \text{ (3)} \quad \neg a \text{ (1)}}{\neg b}}{F}
\end{array}$$

Exercise 4 (First-order formulas). The proofs are not given for all cases. The omitted cases are similar to those already given.

1. $A : \forall x.(P(x) \vee Q(x))$
 $B : (\forall x.P(x)) \vee (\forall x.Q(x))$

(a) $A \models B$: ✗. Consider $(\mathbb{N}, e)$ with

$$\begin{aligned}
e(P) &= \{n \in \mathbb{N} \mid n \bmod 2 = 0\} \\
e(Q) &= \{n \in \mathbb{N} \mid n \bmod 2 = 1\}
\end{aligned}$$

(b) $B \models A$: ✓. Consider any structure (D, e) such that $(D, e) \models B$. We have $\llbracket B \rrbracket_{(D, e)} = 1$. By definition of $\llbracket \cdot \vee \cdot \rrbracket_{(D, e)}$, we know that either $\llbracket \forall x.P(x) \rrbracket_{(D, e)} = 1$ or $\llbracket \forall x.Q(x) \rrbracket_{(D, e)} = 1$. Pick any case, the other is identical. Supposing $\llbracket \forall x.P(x) \rrbracket_{(D, e)} = 1$, we know that for every $d \in D$, $\llbracket P(x) \rrbracket_{(D, e[x:=d])} = 1$. However, this implies that for every $d \in D$, $\llbracket P(x) \vee Q(x) \rrbracket_{(D, e[x:=d])} = 1$, and thus $\llbracket \forall x.(P(x) \vee Q(x)) \rrbracket_{(D, e)} = 1$. Therefore, $(D, e) \models A$ as well.

2. $A : \exists x.(P(x) \vee Q(x))$
 $B : (\exists x.P(x)) \vee (\exists x.Q(x))$

(a) $A \models B$: ✓. Consider any structure (D, e) such that $(D, e) \models A$. We have $\llbracket A \rrbracket_{(D, e)} = 1$. By definition of $\llbracket \cdot \rrbracket_{(D, e)}$, there exists $d \in D$ such that $\llbracket P(x) \vee Q(x) \rrbracket_{(D, e[x:=d])} = 1$. Thus, either $\llbracket P(x) \rrbracket_{(D, e[x:=d])} = 1$ or $\llbracket Q(x) \rrbracket_{(D, e[x:=d])} = 1$. In the first case, $\llbracket \exists x.P(x) \rrbracket_{(D, e)} = 1$, and in the second case, $\llbracket \exists x.Q(x) \rrbracket_{(D, e)} = 1$. In either case, $(D, e) \models B$.

- (b) $B \models A$: ✓. Proof omitted.
3. $A : \forall x.(P(x) \wedge Q(x))$
 $B : (\forall x.P(x)) \wedge (\forall x.Q(x))$
- (a) $A \models B$: ✓. Proof omitted.
- (b) $B \models A$: ✓. Proof omitted.
4. $A : \exists x.(P(x) \wedge Q(x))$
 $B : (\exists x.P(x)) \wedge (\exists x.Q(x))$
- (a) $A \models B$: ✓. Proof omitted.
- (b) $B \models A$: ✗. Consider $(\mathbb{N}, e)$ with

$$\begin{aligned} e(P) &= \{0\} \\ e(Q) &= \{1\} \end{aligned}$$

5. $A : \forall x.(P(x) \rightarrow Q(x))$
 $B : (\forall x.P(x)) \rightarrow (\forall x.Q(x))$
- (a) $A \models B$: ✓. Consider any structure (D, e) such that $(D, e) \models A$. We have $\llbracket A \rrbracket_{(D,e)} = 1$, i.e., for every $d \in D$, $\llbracket P(x) \rightarrow Q(x) \rrbracket_{(D,e[x:=d])} = 1$. Note that $\llbracket P(x) \rightarrow Q(x) \rrbracket_{(D,e[x:=d])} = \llbracket \neg P(x) \vee Q(x) \rrbracket_{(D,e)}$, and thus for each d :

$$\llbracket P(x) \rrbracket_{(D,e[x:=d])} = 0 \text{ or } \llbracket Q(x) \rrbracket_{(D,e[x:=d])} = 1 \quad (1)$$

Assume, towards a contradiction, that $\llbracket B \rrbracket_{(D,e)} = 0$. We must have

$$\llbracket \forall x.P(x) \rrbracket_{(D,e)} = 1 \text{ and } \llbracket \forall x.Q(x) \rrbracket_{(D,e)} = 0 \quad (2)$$

The left conjunct of Equation 2 tells us that for every $d \in D$, $\llbracket P(x) \rrbracket_{(D,e[x:=d])} = 1$. By Equation 1, this implies that for every $d \in D$, $\llbracket Q(x) \rrbracket_{(D,e[x:=d])} = 1$. This contradicts the right conjunct of Equation 2. Therefore, $\llbracket B \rrbracket_{(D,e)} = 1$, and thus $(D, e) \models B$.

- (b) $B \models A$: ✗. Consider $(\mathbb{N}, e)$ with

$$\begin{aligned} e(P) &= \{0\} \\ e(Q) &= \{1\} \end{aligned}$$

6. $A : \exists x.(P(x) \rightarrow Q(x))$
 $B : (\exists x.P(x)) \rightarrow (\exists x.Q(x))$

(a) $A \models B$: ✗. Consider $(\mathbb{N}, e)$ with

$$\begin{aligned} e(P) &= \{0\} \\ e(Q) &= \emptyset \end{aligned}$$

Take $x = 1$ as a witness for A .

(b) $B \models A$: ✓. Consider any structure (D, e) such that $(D, e) \models B$. We have $\llbracket B \rrbracket_{(D,e)} = 1$. As before, by expanding the implication, we must have either $\llbracket \exists x.P(x) \rrbracket_{(D,e)} = 0$ or $\llbracket \exists x.Q(x) \rrbracket_{(D,e)} = 1$.

In the first case, we have that for every $d \in D$, $\llbracket P(x) \rrbracket_{(D,e[x:=d])} = 0$. This implies that for every $d \in D$, $\llbracket P(x) \rightarrow Q(x) \rrbracket_{(D,e[x:=d])} = 1$, and thus in particular $\llbracket \exists x.P(x) \rightarrow Q(x) \rrbracket_{(D,e)} = 1$.

In the second case, there exists $d \in D$ such that $\llbracket Q(x) \rrbracket_{(D,e[x:=d])} = 1$. Applying the interpretation of implication again, we have that $\llbracket P(x) \rightarrow Q(x) \rrbracket_{(D,e[x:=d])} = 1$, and thus $\llbracket \exists x.P(x) \rightarrow Q(x) \rrbracket_{(D,e)} = 1$.

Thus, in either case, $(D, e) \models A$.

Exercise 5 (First-order structures).

- Let $I = (D, \alpha)$ be an interpretation of $\mathcal{L}$, for which $\alpha(B) = b$ and $\alpha(T) = t$. For notational convenience we will denote $\alpha(L)$ by L_α . The consequences of axioms in S are:

$$\begin{aligned} \llbracket \text{Min} \rrbracket_I = 1 & \implies L_\alpha(b, b), L_\alpha(b, u) \text{ and } L_\alpha(b, t) \\ \llbracket \text{Max} \rrbracket_I = 1 & \implies L_\alpha(b, t), L_\alpha(u, t) \text{ and } L_\alpha(t, t) \\ \llbracket \text{Tot} \rrbracket_I = 1 & \implies L_\alpha(u, u) \text{ (by choosing } x = u \text{ and } y = u) \\ \llbracket \text{Tra} \rrbracket_I = 1 & \implies (L_\alpha(t, b) \implies L_\alpha(t, u)) \end{aligned}$$

This greatly reduces the number of choices. We can fill out the partial table for L_α :

L_α	b	u	t
b	1	1	1
u	?	1	1
t	?	?	1

Finally, based on **Tra**, L_α is restricted to be either:

a)	L_α	b	u	t
	b	1	1	1
	u	1	1	1
	t	1	1	1
c)	L_α	b	u	t
	b	1	1	1
	u	0	1	1
	t	0	1	1
b)	L_α	b	u	t
	b	1	1	1
	u	1	1	1
	t	0	0	1
d)	L_α	b	u	t
	b	1	1	1
	u	0	1	1
	t	0	0	1

All the options satisfy reflexivity and transitivity as a consequence of the axioms. However, only $d)$ satisfies the anti symmetry property. Therefore only $d)$ is a partial order.

2.

a)	$E_{\alpha'}$	b	u	t
	b	1	1	1
	u	1	1	1
	t	1	1	1
c)	$E_{\alpha'}$	b	u	t
	b	1	0	0
	u	0	1	1
	t	0	1	1
b)	$E_{\alpha'}$	b	u	t
	b	1	1	0
	u	1	1	0
	t	0	0	1
d)	$E_{\alpha'}$	b	u	t
	b	1	0	0
	u	0	1	0
	t	0	0	1

In all the cases, $E_{\alpha'}$ is an equivalence relation.

3. $E_{\alpha'}$ is always an equivalence relation, whatever is the domain.

Let D' be an arbitrary set and $I' = (D', \alpha')$.

- **Reflexivity:** $\llbracket \text{Both} \rrbracket_{I'} = 1 \implies \forall x \in D'. E_{\alpha'}(x, x) \leftrightarrow L_{\alpha'}(x, x)$
(by choosing $y = x$).

We also know that, $\llbracket \text{Tot} \rrbracket_{I'} = 1 \implies \forall x \in D'. L_{\alpha'}(x, x)$

Therefore $\forall x \in D'. E_{\alpha'}(x, x)$

- **Symmetry:**

$$\llbracket \text{Both} \rrbracket_{I'} = 1 \implies \forall x \in D'. \forall y \in D'. E_{\alpha'}(x, y) \leftrightarrow (L_{\alpha'}(x, y) \wedge L_{\alpha'}(y, x))$$

We also know that for $x, y \in D'$ arbitrary $L_{\alpha'}(x, y) \wedge L_{\alpha'}(y, x) \equiv L_{\alpha'}(y, x) \wedge L_{\alpha'}(x, y)$. Therefore we have that for $\forall x \in D'. \forall y \in D'. E_{\alpha'}(x, y) \leftrightarrow E_{\alpha'}(y, x)$ is a tautology.

- **Transitivity:** $\llbracket \text{Both} \rrbracket_{I'} = 1$ implies that for x, y, z arbitrary in D'

$$E_{\alpha'}(x, y) \leftrightarrow (L_{\alpha'}(x, y) \wedge L_{\alpha'}(y, x))$$

and

$$E_{\alpha'}(y, z) \leftrightarrow (L_{\alpha'}(y, z) \wedge L_{\alpha'}(z, y))$$

Therefore

$$E_{\alpha'}(x, y) \wedge E_{\alpha'}(y, z) \leftrightarrow (L_{\alpha'}(x, y) \wedge L_{\alpha'}(y, z) \wedge L_{\alpha'}(z, y) \wedge L_{\alpha'}(y, x))$$

which implies

$$E_{\alpha'}(x, y) \wedge E_{\alpha'}(y, z) \rightarrow (L_{\alpha'}(x, z) \wedge L_{\alpha'}(z, x))$$

and finally

$$E_{\alpha'}(x, y) \wedge E_{\alpha'}(y, z) \rightarrow E_{\alpha'}(x, z)$$

4. One of the solution consists in choosing $D_1 = D' / E_{\alpha'}$, the set of equivalence classes of $E_{\alpha'}$, $\alpha_1 = (B_1, T_1, L_1)$ where $B_1 = [b]_{E_{\alpha'}}$, $T_1 = [t]_{E_{\alpha'}}$ and

$$L_1 := \{([x]_{E_{\alpha'}}, [y]_{E_{\alpha'}}) : L_{\alpha'}(x, y) \text{ where } x, y \in D'\}$$

Reflexivity and transitivity are immediate as they have been proven for $L_{\alpha'}$.

Antisymmetry comes from the fact that for arbitrary $[x]_{E_{\alpha'}}, [y]_{E_{\alpha'}} \in D_1$

$$\begin{aligned} & L_1([x]_{E_{\alpha'}}, [y]_{E_{\alpha'}}) \wedge L_1([y]_{E_{\alpha'}}, [x]_{E_{\alpha'}}) \\ \implies & L_{\alpha'}(x, y) \wedge L_{\alpha'}(y, x) \\ \implies & E_{\alpha'}(x, y) \\ \implies & [x]_{E_{\alpha'}} = [y]_{E_{\alpha'}} \end{aligned}$$

Therefore L_1 is a partial order.